



Lost Pet Tracker

AI-Assisted Matching for Lost and Found Pets

Benjamin Biscomb

Nimra Shakoor

Amari Harrington

Khasim Raza

Benjamin Biscomb

bbiscom1@students.towson.edu <https://github.com/vivi-in-4D>

EDUCATION

Towson University August 2026 - Present
M.S. in *Computer Science*

Louisiana Tech University August 2020 - May 2026
B.S. in Cyber Engineering, Minor in Computer Science

RELEVANT PROFESSIONAL EXPERIENCE

Load-Perfect October 2025 - Present
System Administrator

- Develop and maintain a full stack web application supporting business operations for a trucking company
- Host the site on a DigitalOcean droplet running headless Ubuntu, with file upload, file-type checking, and ClamAV malware detection
- Integrate the Gmail API to deliver uploaded files to company addresses from the droplet
- Administer company email through Google Workspace

Louisiana Tech University January 2023 - June 2023
Software Developer

- Developed a Department of Defense funded CTF using Kubernetes clusters to rapidly spin up Kali Linux containers in high volume, designed to teach students red-team skills
- Built a Kubernetes cluster capable of rapid spin-up of 100+ Linux containers simultaneously
- Contributed front-end web development and testing for the CTF platform

TECHNICAL SKILLS

- Programming languages: C, C++, Java, Python, C#, Rust, Bash
- Web and Frontend: JavaScript, HTML, CSS, JSON
- Assembly/ISAs: x86, MIPS
- Hardware Development: VHDL, Verilog, Microblaze, Vivado, Xilinx
- Embedded Systems: ARM, Pico, UART
- Systems and Infrastructure: Linux, VMs, containers, DNS, DHCP, SFTP, SQL, AD/DC
- Security and Offensive Tooling: NIST SP 800-171 Rev. 3, firewalls, access control, Metasploit, Nmap, Wireshark
- AI and ML: Claude Code, GANs, PyTorch, StyleGAN2
- DevOps and Version Control: Git, Docker, Kubernetes
- Reverse Engineering: Ghidra, JADX
- Microsoft Office Certifications: Excel, Word, Powerpoint, Outlook Access

NOTABLE PROJECTS

Contract Compliance Monitor (Senior Capstone) August 2024 - May 2025
Project Lead

- Built a network-wide security tool designed to enforce compliance with NIST SP 800-171 Rev. 3
- Integrated file tracking, database management, AD/DC, encryption at rest, and SFTP with a web UI
- Led a team of 4, assigning work, determining objectives, and overseeing high-level system design
- Designed the SFTP, encryption, and web UI components, and authored all technical documentation

Nimra Shakoor

nshakoor1@students.towson.edu <https://github.com/nshakoor18>

EDUCATION

Towson University

M.S. in *Computer Science*

B.S. in *Information Technology*

August 2026 - Present

August 2025

CERTIFICATIONS

- DHS Trusted Tester for Web
- Google UX Design Certificate

RELEVANT PROFESSIONAL EXPERIENCE

Independent Web Development & Accessibility Work

2025 - Present

Web Developer / Accessibility Specialist

- Develop and evaluate websites with an emphasis on accessibility, usability, and responsive design
- Perform accessibility testing using NVDA, ANDI, Axe DevTools, WAVE, and keyboard-only navigation
- Identify and remediate accessibility issues involving semantic structure, forms, focus management, alternative text, ARIA, and keyboard interaction
- Apply WCAG and Section 508 requirements during design, development, and testing

TECHNICAL SKILLS

- Programming and Web: C++, HTML5, CSS3, JavaScript, Bootstrap 5
- Accessibility: WCAG, Section 508, WAI-ARIA, NVDA, ANDI, Axe DevTools, WAVE
- UX/UI: Figma, usability testing, accessibility evaluation, responsive design
- Development Tools: Git/GitHub, VS Code, Netlify
- Web Development: Form validation, semantic HTML, focus management, responsive interfaces, JSON-LD

NOTABLE PROJECTS

Fast Towing & Recovery Business Website

July 2025 - Present

Web Developer / Accessibility Specialist

- Designed and developed a responsive multi-page business website using HTML, CSS, JavaScript, and Bootstrap 5
- Implemented accessible forms, keyboard navigation, focus management, semantic HTML, ARIA, and accessible interactive components
- Conducted accessibility testing using NVDA, ANDI, Axe DevTools, WAVE, and keyboard-only navigation
- Implemented client-side form validation, accessible error handling, structured metadata, and responsive layouts

Delta vs. American Airlines Usability Study

January 2025 - May 2025

Human-Computer Interaction Project

- Conducted a comparative usability study involving 20 participants
- Evaluated user interaction and usability across comparable tasks on the Delta and American Airlines websites
- Analyzed participant results to identify usability differences and opportunities for interface improvement

Amari Harrington

aharrin7@students.towson.edu <https://github.com/AmariVashon>

EDUCATION

Towson University

August 2026 - Present

Master's, *Applied Information Technology*

University of Maryland, College Park

August 2022 - May 2026

Bachelor's of Science, Information Science

- Gemstone Honors College

Relevant Coursework

Object-Oriented Programming, Calculus II, Linear Algebra, Statistics, Algorithms, Computer Systems, Discrete Mathematics, Networking, Web Development

RELEVANT PROFESSIONAL EXPERIENCE

National Aeronautics and Space Administration (NASA)

June 2022 - August 2022

Data Science Intern

- Developed a Python and PyQt5-based Graphical User Interface (GUI) to oversee real-time movement and positioning of a robot by using a USB/serial connection to communicate with a Teensy 4.0
- Documented development of the application and presented results to multiple colleagues and mentors

National Oceanic and Atmospheric Administration (NOAA)

August 2021 - May 2022

Data Science Intern

- Designed an algorithm using Python that traversed thousands of longitudinal and latitudinal positions to determine the distance between a Synthetic Aperture Radar's detected oil platform and actual oil platform position

TECHNICAL SKILLS

Python, Java, C, JavaScript, HTML/CSS, Git, OCaml, SQL, PacketTracer

NOTABLE PROJECTS

NBA Injury Predictor

Summer 2025

- Built an end-to-end ETL pipeline to ingest, clean, and transform NBA player and game data for modeling
- Trained a binary decision tree classifier to predict player injury risk over a 7-day forecasting window
- Engineered features related to workload, injury history, and player statistics to improve model accuracy

AI In Education

August 2024 - May 2026

- Collaborated with a 6-person research team to examine how middle school teachers and collegiate professors use and perceive AI tools in classroom instruction
- Analyzed qualitative and quantitative data to identify trends and understand AI adoption in the classroom

Khasim Raza

klnu4@students.towson.edu <https://github.com/Khasimraza/>

EDUCATION

Towson University <i>M.S. in Computer Science</i>	August 2025 - Present
Acharya Institute of Technology, Bangalore <i>B.E. in Electrical & Electronics</i>	2016 - 2020

CERTIFICATIONS

- Microsoft Azure Administrator (AZ-104)
- Azure Solutions Architect Expert (AZ-305)

RELEVANT PROFESSIONAL EXPERIENCE

Microland - General Electric Project <i>Cloud Infrastructure Engineer</i>	March 2021 - December 2024
• Operated endpoint infrastructure for 500,000+ devices across 46 countries; handled 15,000+ monthly requests on a follow-the-sun on-call rotation • Built a Python/PowerShell automation library adopted by a 12-engineer team, eliminating 15+ hours/week of manual work	

HACKATHON MENTORSHIP & JUDGING

CMU Hackathon, Carnegie Mellon University <i>Judge & Mentor</i>	2026
• Advised teams on cloud deployment and getting prototypes hosted under time pressure	
LA Hacks, University of California, Los Angeles <i>Cloud Infrastructure Engineer</i>	2026
• Guided 1,000+ participants through deployment, debugging, cloud services, API integration, and scoping decisions	

TECHNICAL SKILLS

- Cloud & Infrastructure: Microsoft Azure, AWS (EC2, S3, RDS), GCP, Linux, Multi-Cloud Architecture, VM Migration, FinOps
- Programming & Automation: Python, PowerShell, Bash, SQL, JavaScript, Git, Microsoft Graph API, Azure CLI, CI/CD Pipelines
- Platform & Identity: Microsoft Intune, Azure AD / Entra ID, Conditional Access, Apple DEP & ABM
- Data & ML: ETL Pipelines, Kafka, scikit-learn, ServiceNow (ITSM/CMDB), Power BI, Tableau
- DevOps & IaC: Terraform, Ansible, Docker, GitHub Actions, Kubernetes (in progress)

NOTABLE PROJECTS

Real-Time Infrastructure Monitoring Platform
• Event-driven pipeline ingesting 500K+ device health events/day across 50+ metrics; seasonal baselining caught degradation before user impact, cutting MTTR 45%
Automated Incident Analytics Engine
• Triage classifier trained on 100K+ tickets across 12 categories, reaching 85% accuracy vs. a 61% baseline; served via REST API with weekly retraining

Planning and Scheduling

Assignee Name	Email	Task(s)	Duration	Dependency	Due Dates
Benjamin Biscomb (Coordinator)	bbiscom1@students.towson.edu	Github Creation Kanban Creation Technical Writing Cover Page Schedule Creation Resume	10 hours	Project Topic System Requirements	09/22/2026 09/25/2026 09/28/2026
Nimra Shakoor	nshakoo1@students.towson.edu	Project Topic Problem statement Resume	6 hours	None	09/22/2026 09/25/2026
Amari Harrington	aharrin7@students.towson.edu	System Requirements Context Model Resume	2 hours	None	09/22/2026 09/25/2026 09/28/2026
Khasim Raza	klnu4@students.towson.edu	Resume Partial Contribution	2 hours	Information from Coordinator	09/25/2026 09/28/2026

Problem Statement

Product Overview

The Lost Pet Tracker is an interactive application designed to help pet owners and community members locate and identify lost pets. Users will be able to create two types of reports: lost pet reports and found pet reports. Reports contain photographs, physical descriptions, distinctive characteristics, dates, and either the location the pet was last seen or the location the pet was found, depending on the report type. These reports will be stored in a searchable database and analyzed to identify potential matches.

The Lost Pet Tracker will use an AI-based matching process; this process will consider lost and found reports that have locations within an Administrator-configured radius of each other. Location reporting will use Google Maps API. The use of Artificial Intelligence will allow for each post's data to be considered holistically, regardless of semantic inconsistencies. The factors used by the matching process may include photographs, species, estimated breed or breed group, size, color, distinctive markings, geographic proximity, and the dates associated with the reports. Image analysis may also suggest a likely breed or breed group when a user is unsure, while allowing the suggestion to be reviewed or corrected. When the Lost Pet Tracker determines two posts have enough matching characteristics, their posters will be notified via email. At any time after post creation, its poster may delete the post. It is recommended that all posts are deleted after a pet has successfully been returned. To maintain user privacy, posts will automatically be deleted after 3 months of poster inactivity.

Currently, information that could help reunite a lost pet with its owner may be distributed across social media, neighborhood groups, animal shelters, veterinary offices, flyers, and lost-and-found databases. Differences in how people describe the same animal can make searching more difficult. The Lost Pet Tracker will provide a structured system for connecting this information and presenting potential matches for users to review. AI will assist this process, but the final determination of whether two reports represent the same animal will remain with the users.

Existing lost-pet services provide different approaches to pet recovery. Petco Love Lost provides a streamlined reporting process that uses location information and pet photographs to search for potential matches. Pet FBI maintains a searchable national lost-and-found database and automatically suggests potential matches based on factors such as location, date, and animal type. PawBoost emphasizes community outreach and geographic search and is also testing its Rescue Squad feature, which presents users with potential lost-and-found matches for human review. The proposed Lost Pet Tracker will build on these existing approaches while focusing on explainable, multi-factor matching. The system will combine available information such as photographs, reported physical characteristics, distinctive markings, geographic proximity, and the time between lost and found reports. When possible, the interface will show the characteristics that contributed to a suggested match, allowing users to understand and evaluate the recommendation rather than relying solely on an automated result. The project will also emphasize a low-friction reporting and comparison workflow. Users will be able to provide structured pet information and review potential matches through an interface designed to communicate important characteristics through both images and text. AI will perform the matching process and the Lost Pet Tracker will send recommendations to the user.

Objectives and Project Scope

The primary objective of the project is to develop a functional system in which users can create, search, update, and resolve lost and found pet reports, while receiving AI-assisted notifications for potential matches. Core activities will include submitting report information and photographs, searching stored reports, reviewing suggested matches, and updating a report when an animal has been located.

Accessibility will be incorporated in all areas of the project. Core interactions will support keyboard navigation and screen readers through accessible form labels, controls, error messages, and status updates. Information presented visually through photographs, maps, color, or AI-generated match results will also include appropriate textual or nonvisual alternatives. In particular, users should be able to understand why two reports are considered a potential match without relying solely on visual comparison of the pet photographs.

This project will prioritize the core reporting and AI-assisted matching workflow. Additional capabilities, such as an in-application chat feature and AI-assisted moderation may be added if time and resources permit. Keeping these features outside the initial requirements will allow our team to focus on producing a functional and testable system within the given timeframe.

Feasibility and Technical Interest

The project will be developed with the intent to host it on our own machine, but could easily be ported to a cloud hosting provider such as DigitalOcean. Our host device will be identified during development. We currently plan to integrate three locally run and pretrained AI models; our model choice might update throughout the project if our current choices do not fit our constraints. Currently, DINOv3 will be used to determine image similarity; Qwen3-VL will be used for attribute extraction; and all-MiniLM-L6-v2 will be used for text description matching. A web-based interface will communicate with a SQL database containing lost and found pet reports, account information, photographs, locations, and report statuses.

The technical work will focus on integrating AI-review processes with structured report data and developing logic for identifying and ranking potential matches. Additionally, much of the development process will be spent on backend and frontend web development. However, we will utilize premade resources and libraries where possible. This approach makes the project achievable with the team's available time and computing resources while still requiring software engineering, integration, and testing.

The project presents an additional technical challenge because user-submitted information may be incomplete, inconsistent, or incorrect; two users may give very different descriptions to the same animal. Normal rule-based matching frameworks, such as those that take in structured data and use regular expressions, would struggle with this inconsistency. Our holistic AI review approach is hardened against semantic inconsistencies; but it introduces other modes of failure such as misreading low quality images. For usability, our system must address imperfect sources of information by communicating why a potential match was suggested and allow users to contact each other without the AI-system determining a match.

The completed system can be tested using sample lost and found pet reports to evaluate report creation, AI-assisted analysis, potential-match generation, accessibility, and the process of resolving a report. The combination of database design, account management, AI integration, geographic information, web development and accessibility makes this project substantial enough for a semester-long team project while remaining interactive and testable.

System Requirements

Components

The Lost Pet Tracker will consist of the following components/modules:

1. User Account and Post Handling
2. Front-End User Interface
3. Web Backend
4. AI-Based Recommendation and Matching
5. Email communication
6. Two-Factor Code Generation
7. Google Maps Integration
8. SQL Database

Component Definition

At a high level, these will operate as follows:

1. User Account and Post Handling

The Lost Pet Tracker does not have a standard account creation process. When creating a post, users will be given a private account number unique to their post. This account number will be sent to them via email, a required field when creating a post. Users must keep track of their own account number. When creating a post, users must click a verification link sent to their email before it is displayed publicly and used for matching. If a user does not verify their email, the post will be deleted 24 hours later. Users may log into their account to make changes to their post or delete it. Upon deletion of a post, all data related to the post, including a user's account number and contact information will be deleted. To log into an account users must provide an account number and two-factor authentication code sent to their associated email. During post creation, users can provide one or more forms of contact information, which can be either an email address or phone number. Users can mark each piece of contact information as "public", "email-only", or "private". "Public" contact information is displayed on a user's public post, which users will be made aware of during post creation. Users who have received notification of a match will be given the other poster's contact info marked "public" and "email-only" in their notification email. After three months of account inactivity, the account and post will be deleted. Posts are divided into two categories: lost pet reports and found pet reports.

2. Front-End User Interface

The web interface will handle account creation, account changes, post creation, post changes, post deletion, and a location-integrated search feature.

3. Web Backend

The backend will interact with the front-end and manage interactions between system components. This includes listening for user events, running AI-assisted matching, sending emails, account management, database management, and serving data.

4. AI-Based Recommendation and Matching

The Pet Tracker will use Artificial Intelligence to match lost and found pet reports in three ways. Currently, DINOv3 will be used to determine image similarity; Qwen3-VL will be used for attribute extraction; and all-MiniLM-L6-v2 will be used for text description matching. Our model choice might update throughout the project if our current choices do not fit our constraints.

5. Email communication

User posts require at least one email address for account creation. Before a post is made, the email address will be verified through a single-use timed two-factor authentication code. Users can choose to expose or hide their email in their post, but at least one communication method must be exposed. When our AI-matching process identifies a match, a recommendation email will be sent to both parties.

6. Two-Factor Code Generation

The Pet Tracker will use a simple Account Number + System Time + Salt two-factor code generation. This code will be sent via email.

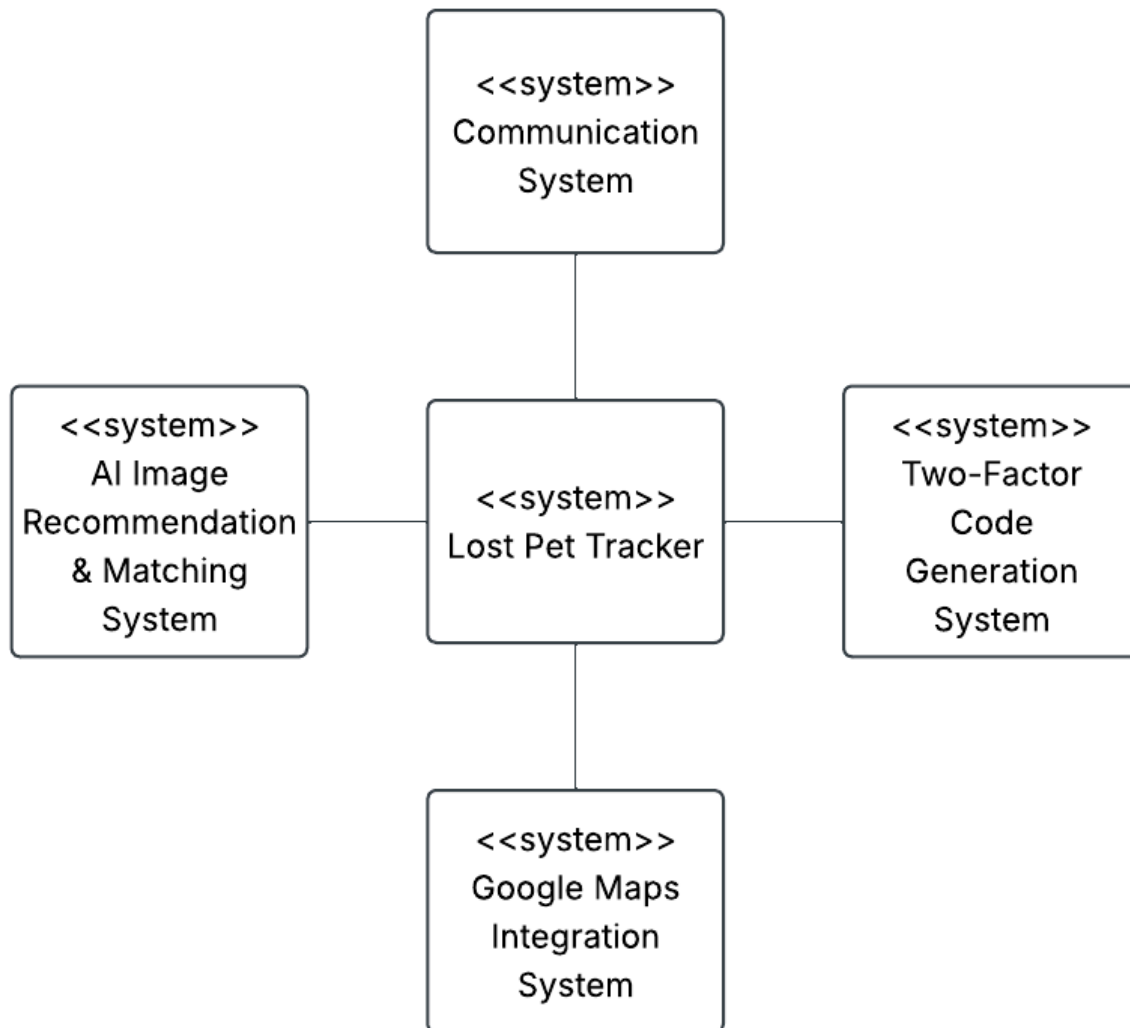
7. Google Maps Integration

The Pet Tracker will integrate google maps API to give a useful graphical map interface to users searching the website. Additionally, the API will be used in post creation to give accurate coordinates and regions for pet locations. These coordinates will be stored in the Pet Tracker's database.

8. SQL Database

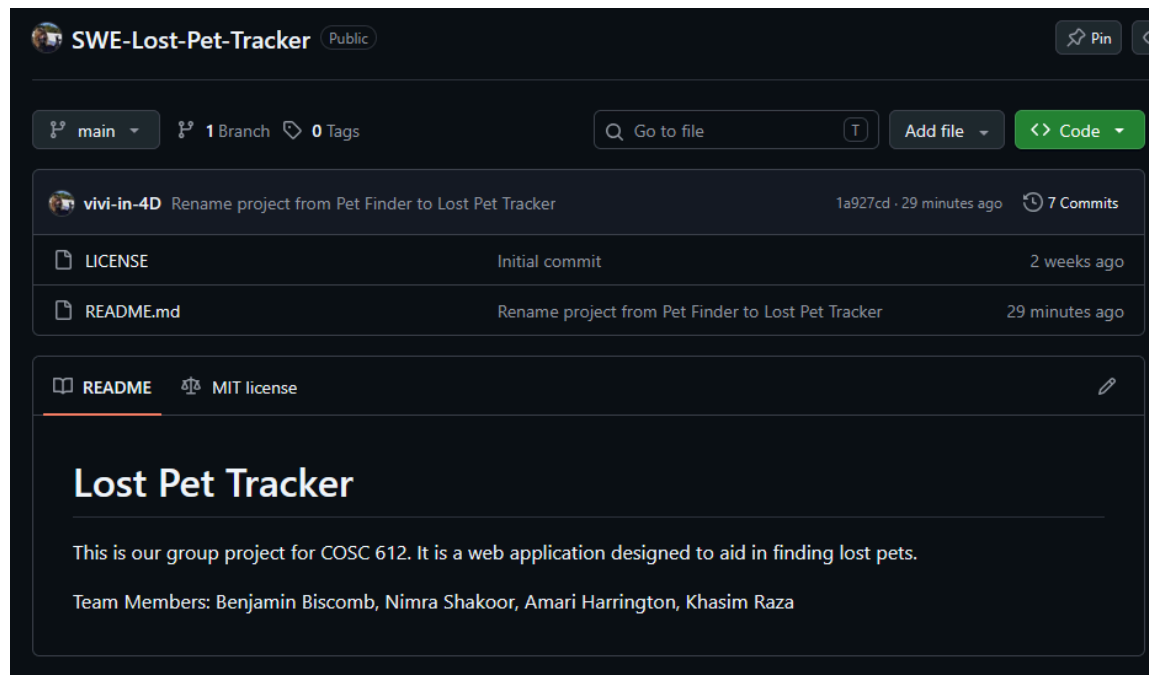
The Pet Tracker database will contain a table for posts, which will store post ID, encrypted account numbers, location information, images, post descriptions, a post type boolean, and a post verified boolean. This table will also contain fields for AI-generated descriptions and metadata. A second table will be used to store information related to communication and recommendations; this table will store matches using post ID, a distance between posts and an already contacted boolean.

Context Model



Appendix

Readme and [Github Repository](#) Front Page:



[Kanban Board:](#)

